

TalentPulse AI

Master Project Plan: Autonomous Enterprise HR Recruitment & Lifecycle Platform

Version: 2.0 (Enterprise Release)

Status: Active Architecture & Development

Core Framework: Next.js 16 + FastAPI + LangGraph

Security: SOC-2 Type II & HITL Guardrails

1. Executive Summary & Problem Statement

TalentPulse AI is an enterprise-grade autonomous AI recruitment platform designed to eliminate recruiter bottlenecks while maintaining strict **Human-In-The-Loop (HITL)** compliance. Modern recruitment operations are heavily fragmented across disconnected tools (ATS, email clients, calendars, LinkedIn, document signing, and background check portals).

Key Industry Pain Points Addressed:

- **Recruiter Fatigue:** 65% of recruiter time is consumed by low-leverage coordination (scheduling, scorecard tallying, drafting postings).
- **Extended Time-to-Hire:** Traditional recruitment takes 36 to 48 days, causing high drop-off among top candidates.
- **Compensation & Parity Risks:** Manual salary negotiations frequently breach internal salary equity bands and equal pay regulations.
- **Prompt Injection & Untrusted Data:** Adversarial resume submissions attempting to trick ranking models are mitigated via vision OCR and strict input sanitization.

2. End-to-End 10-Stage Recruitment Lifecycle

TalentPulse AI covers the complete talent acquisition lifecycle across 10 unified stages:

Stage	Operational Scope	AI Agent Functionality	HITL Gate
1. Requisition	Role opening & JD formulation	Generates role parameters, salary bands (\$140k-\$165k), and skill matrices	No
2. Social Sourcing	LinkedIn campaign dispatch	Drafts social posts, tracks comments, and imports inbound profile leads	Yes (Publish)
3. Ingestion & OCR	Resume parsing & sanitization	Multi-column OCR via PyMuPDF + Vision LLM with prompt injection firewall	No (Async)
4. Hybrid Match	Candidate ranking & discovery	Fuses ChromaDB dense vectors with SQL filters; grounded source citations	No (Read)
5. Scheduling	Interview coordination & conflicts	Automated Google Calendar booking & autonomous reschedule conflict negotiation	Yes (Send)
6. Scorecards	Panel debrief & synthesis	Aggregates ratings into an AI consensus debrief (Strong Hire / Hire / No Hire)	Yes (Advance)
7. Negotiation	Candidate counter-offer analysis	Validates asks against budget band; models sign-on/equity counter-packages	Yes (Approve)
8. Offer Letters	Formal employment contract	Generates binding offer letters with 4-year vesting and signature placeholders	Yes (Dispatch)

9. BGV Clearance	Background verification & KYC	Tracks 4 checks (Employment, Degrees, Criminal, Sanctions) with risk scoring	Yes (Clear)
10. Welcome Kit	Day 1 onboarding preparation	Dispatches orientation itinerary, reporting contact, and KYC document checklist	Yes (Dispatch)

3. Technical Architecture & Component Specifications

Frontend Layer: Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS v4, Lucide Icons, and React-Markdown. Implements responsive sidebars, interactive HITL approval queues, real-time streaming chat, candidate pipeline Kanban, and activity audit trails.

Backend Core: FastAPI (Python 3.13), async SQLAlchemy 2.0, Alembic migrations, Pydantic v2 validation, and SSE streaming. Features Scoped MCP tool definitions ensuring least-privilege agent execution.

Agent Orchestration: LangGraph multi-agent state machines with ReAct tool-calling loops, session checkpointers, and prompt-injection firewalls.

Data & Vector Persistence: PostgreSQL 16 for relational models (Candidates, Scorecards, Negotiations, Offers, Logs) + ChromaDB for dense semantic vector search (768/1536-dim embeddings).

Component	Technology / Framework	Primary Purpose
UI & Experience	Next.js 16, React 19, Tailwind CSS	Interactive recruiter dashboard, chat stream & HITL queue
API Gateway	FastAPI, Uvicorn, SSE / WebSockets	High-throughput async endpoint routing & streaming
Agent Runtime	LangGraph, Claude 3.7 Sonnet / GPT-4o	Autonomous multi-agent lifecycle planning & tool calling
Relational Store	PostgreSQL 16, Asyncpg, SQLAlchemy	ACID transaction logs, candidate records, audit ledger
Vector Store	ChromaDB / pgvector	Dense resume chunk embeddings & cosine similarity
Async Worker	Celery, Redis 7	Background PDF OCR, parsing & inbound email processing
Security	OAuth2, JWT, AES-256, Prompt Guard	PII masking, SOC-2 encryption & adversarial defense

4. Installation & Environment Setup Guide

4.1 Frontend Setup (Next.js 16)

```
# 1. Navigate to frontend & install dependencies
cd frontend
npm install

# 2. Configure environment variables in .env.local
NEXT_PUBLIC_API_BASE_URL=http://localhost:8000/api/v1

# 3. Start development server
npm run dev # Accessible at http://localhost:3000

# 4. Production build verification
npm run build
```

4.2 Backend Setup (FastAPI & LangGraph)

```
# 1. Navigate to backend & create virtual environment
cd backend
python -m venv .venv
.\.venv\Scripts\Activate.ps1 # Windows (or 'source .venv/bin/activate' on Linux)

# 2. Install dependencies
pip install -r requirements.txt

# 3. Start FastAPI application
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
# Interactive Swagger Docs at http://localhost:8000/docs
```

5. Future Goals & Strategic Roadmap (2026 – 2027)

Milestone & Phase	Target Window	Key Deliverables & Enterprise Capabilities
Phase 1: Core Platform	Q1 2026 (Completed)	Next.js 16 workspace, 10 HITL approval cards, Audit timeline, Requisition generator, Pipeline Kanban.
Phase 2: Backend Core	Q2 2026 (In Progress)	FastAPI LangGraph multi-agent loop, PyMuPDF OCR async worker, ChromaDB hybrid search, prompt injection guard.
Phase 3: Live Connectors	Q3 2026	LinkedIn Graph API live posting, Google & Microsoft Calendar 2-way sync, SendGrid SMTP deliverability tracking.
Phase 4: Multi-Tenant SaaS	Q4 2026	Enterprise RBAC, Okta/SAML SSO, custom department salary bands, regional compliance auditing.
Phase 5: Voice AI Screener	Q1 2027	Autonomous 10-minute real-time WebRTC AI phone screening agent with auto-scorecard compilation.
Phase 6: ATS Ecosystem	Q2 2027	Bi-directional sync connectors for Workday, Greenhouse, Lever, and SAP SuccessFactors.

6. Quality Benchmarks & SLA Targets

- **Retrieval Precision@5:** 90%+ accuracy on benchmark queries matching structured criteria.
- **Hallucination Rate:** 0.0% ungrounded claims regarding candidate qualifications (all citations verified).
- **HITL Safety Interception:** 100.0% of external state-changing actions require human authorization.
- **Injection Defense:** Zero ranking skew or execution hijack from adversarial prompt injection in resumes.
- **Streaming Latency:** First token response ≤ 600ms; complex multi-agent execution ≤ 3.5s.

Document Sign-off: Architecture reviewed and verified for Enterprise Deployment. Refer to *project_architecture.pdf* for complete system topology and workflow diagrams.